

Return Momentum and Global Portfolio Allocations

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Abstract

In this paper, we report the first empirical tests concerning the international investment strategies of a panel of investment houses from 1982 through 1999. The previously untapped data for this study comes from surveys published in the *Financial Report*, a confidential newsletter purchased by *The Economist Newspaper, Ltd.*, in 1989. In the surveys, the investment houses recommend portfolio allocations among equity, bonds, and cash, as well as equity holdings across six country categories.

We test whether changes in these portfolio weights are consistent with the notion of momentum trading. That is, we test whether the recommended weights change in response to relative asset return performance in previous periods. We find strong evidence of momentum trading for strategic asset allocations to equities and cash as well as strong evidence of momentum trading for country-specific, or tactical, equity allocations. In addition, we find the results differ conditional upon the recommended weights to bonds. Specifically, investment houses that recommend a lower percentage to bonds, on average, appear to be more likely to engage in momentum trading for equities. We also test whether changes in portfolio weights are consistent with the notion of two alternate investment strategies. We find no evidence that changes in portfolio weights are consistent with changes in value investing variables or with changes in macroeconomic variables.

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Assessing the value of investment advice is a compelling inquiry. Indeed, the pursuit of this question is the focus of some significant recent research concerning financial markets in the United States. However, to date, no study specifically examines investment advisor recommendations concerning strategic and tactical asset allocations for international financial markets.

In this paper, we ask: Are changes in portfolio recommendations related to the relative historic return performance among assets in the investment opportunity set? That is, our focus is on testing whether individual investment houses modify recommended portfolio weights in the hope of benefitting from the continued performance, or “momentum,” of asset returns.

We examine the recommended international investment strategies of a panel of investment houses from April 1982 through October 1999. The recommended portfolio weights come from surveys published every six weeks beginning on April 8, 1982 in the *Financial Report*, a confidential newsletter purchased by *The Economist Newspaper, Ltd.*, in 1989. *The Economist* continued to conduct the survey but, starting on March 25, 1989, published it every three months.

In the survey, international money managers located in different countries were asked to provide asset allocation recommendations for a hypothetical investor who has “no existing investments, no overriding currency considerations, and the investment objective of long-term capital growth.” Throughout most of the sample, the investment houses report their recommended portfolio allocations among equity, bonds and cash, as well as equity holdings by country for the United States, Japan, the United Kingdom, Germany, France, and “Other.”¹

We find strong evidence that, as a group, the surveyed investment houses (the “Houses”) adjusted their strategic allocations to equities and cash in response to recent returns in these broad asset classes. We also find strong evidence that the Houses, as a group, change their tactical allocations across country-specific equity weights in response to relative return rankings across countries. These findings are consistent with the notion that the Houses engaged in momentum trading.

In contrast to the strong evidence concerning equities and cash, however, there is no evidence

¹The *Financial Report* christened these recommended portfolio weights, “The Perfect Portfolio.”

that the Houses used a momentum trading strategy for bonds. To further investigate this finding, we sub-divided the sample into two groups as follows. At each recommendation date, we compute the average recommended bond holdings across all Houses. Then, we define “High-Bond Houses” to be those that recommend bond holdings at a level greater than the mean and define “Low-Bond Houses” to be those that recommend bond holdings at a level less than the mean.

We find evidence consistent with the notion that the Low-Bond Houses used momentum strategies for equities. For cash, we also find evidence consistent with the notion of momentum trading for both High-Bond and Low-Bond Houses. For bonds, we find no evidence of momentum trading. However, we do find evidence that High-Bond and Low-Bond Houses responded in a statistically different manner concerning their recommended bond allocations.

We also investigate the possibility that the Houses follow two alternative investment strategies for equities. Specifically, we consider the possibilities that the Houses employ a value-investing approach involving Price-Earnings Ratios and Dividend Yields, or employ an investment approach based on changes in macroeconomic variables.

While we find evidence that the Houses revised their recommended strategic equity allocations in response to changes in the Price-Earnings Ratio and the Dividend Yield, the nature of our findings suggest that these changes are not necessarily consistent with a value approach. Instead, our findings buttress an argument for momentum investing.

The only macroeconomic variable with a significant relation to subsequent changes in strategic equity allocation recommendations was inflation. Test statistics for Industrial Production growth and U.S. Dollar Trade-Weighted growth are insignificant. Finally, there is almost no evidence that the Houses used a macroeconomic investing approach for tactical, or country-specific, equity allocations.

The paper is organized as follows. Following a brief review of related literature in Section 1, Section 2 presents a description of the data used in the study. Selected summary statistics for the sample are discussed in Section 3. A description of the test statistics used to measure momentum trading appears in Section 4. In Section 5, we present empirical test results. An examination of alternative investment strategies appears in Section 6. Section 7 is a summary and conclusion.

1 Related Research.

1.1 Recent Evidence on Stock Return Momentum.

Several recent studies investigate whether return momentum exists in stock returns, and conclude that average stock returns are related to past performance. For example, Jegadeesh and Titman (1993) show that over an investment horizon of three to twelve months in U.S. equity markets, past “winners,” on average, continue to outperform past “losers.” Evidence of momentum is not limited to U.S. equity markets. Rouwenhorst (1998) finds that momentum strategies are profitable in 12 European equity markets. Rouwenhorst (1999) further reports that emerging equity markets exhibit momentum. Chan, Hameed, and Tong (2000) find evidence of profitability for momentum strategies implemented on international stock market indexes.

1.2 Studies of Momentum Capturing Investment Strategies.

Investors following the basic momentum strategy of buying past winners and selling past losers have been named “positive feedback traders” [De Long, Shleifer, Summers, and Waldmann (1990)]. Nofsinger and Sias (1999) conclude that institutional investors buy past winners and sell past losers. Some studies, however, uncover evidence of asymmetric momentum. Lakonishok, Shleifer, Thaler, and Vishny (1991) find that pension fund managers buy past winners but do not sell past losers. In addition, Wermers (1999) suggests that mutual funds buy past winners, but finds little evidence that mutual funds sell past losers.

A host of subsequent studies by financial economists show that professional investors implement various strategies in an attempt to exploit momentum. Grinblatt, Titman, and Wermers (1995) show that mutual fund managers tend to pursue momentum strategies. Badrinath and Wahal (2000) find weak evidence of momentum strategies for the more general category of 13-F filers.²

Other studies extend the investigation to individual investors, but the results are far from uniform. Ippolito (1992) and Sirri and Tufano (1998) present evidence that individual and institutional

²Section 13-F of the 1934 Securities and Exchange Act mandates that entities with discretionary trading authority for common stock, preferred stock, and convertible bonds in portfolios of over \$100 million file a 13-F report detailing their holdings.

investors extrapolate past mutual fund performance. Del Guercio and Tkac (2000) show that mutual fund investors flock to recent winners but do not withdraw assets from recent losers. Bange (2000) finds that small investors buy and sell stocks as if past market movements can be extrapolated into the future. Evidence of momentum trading is also found in published investment advice. Jaffe and Mahoney (1999) report that investment newsletters tend to recommend purchasing securities that have performed well in the past.

Several studies extend the momentum question to international markets. Brennan and Cao (1997) present a theoretical model and empirical evidence that support the view that foreign investors should pursue momentum strategies because they are less informed than domestic investors. Froot, O’Connell and Seasholes (2001) and Choe, Kho and Stulz (1999) find that foreign investors tend to be momentum traders. Using unique Finnish data, Grinblatt and Keloharju (2000, 2001) show that foreign investors pursue momentum strategies.³

2 Data Set Construction.

The raw data for this paper comes from three sources. The first source consists of the asset allocation recommendations from the Houses. The second set consists of daily equity, bond, and cash returns. Equity returns are further broken down into country returns for the U.S., Japan, the U.K., Germany, France, and “Other.” The first two data sets are described in more detail below.

The third set consists of macroeconomic variables used in tests of alternate investment strategies. Specifically, these variables include the Price-Earnings Ratio and Dividend Yield on the MSCI World Index, as well as Industrial Production, Inflation, and the U.S. Dollar Trade-Weighted Index. In

³A separate thread of research centers on the question of whether “experts” possess superior timing ability. That is, is it possible for the experts to make periodic changes in asset allocations so that recommended portfolios outperform benchmark portfolios? Empirical results are divided by the periodicity of the sample. Examining monthly portfolio recommendations in U.S. markets, Metrick (1999), Jaffe and Mahoney (1999), and Graham and Harvey (1996) conclude that investment newsletters fail to offer superior market timing advice. However, studies of the actual performance of mutual fund managers [Bollen and Busse (2001)] and professional market timers [Chance and Hemler (2001)] demonstrate successful market timing when daily data, but not monthly data, are used. It is possible that the Houses in our sample possess timing ability. However, in this study, we use recommendations that are published more than a month apart. Even if the Houses possess market timing ability, the results of previous research suggest that studying market timing ability using this sample is unlikely to reveal superior timing ability. Consequently, we do not investigate market timing as a possible motivation for recommended changes in portfolio allocations.

addition, we obtained Price-Earning Ratios, Dividend Yields, Industrial Production, Inflation and U.S. Dollar Exchange Rates by country. The Industrial Production and Inflation are from the *OECD Main Economic Indicators*, and the MSCI data comes from Datastream©. Exchange rate data for the U.S. Dollar Trade-Weighted index and country-specific U.S. Dollar Exchange Rates come from the Federal Reserve Economic Database, FRED©.

2.1 Asset Allocation Recommendations.

The data for this study comes from surveys published in the *Financial Report*, a confidential newsletter purchased by *The Economist Newspaper, Ltd.*, in 1989. Beginning on April 8, 1982, the *Financial Report* published its survey every six weeks. *The Economist* continued the survey, but beginning on March 25, 1989, published it approximately every three months.

According to the staff of *The Economist*, the survey is currently conducted via e-mail. Research analysts at each House complete the survey. The survey is conducted on the Wednesday before the Friday publication date of *The Economist*. In the early days of the sample, however, data-gathering and printing technology might not have allowed for such a quick turn-around. In addition, it is not known exactly when each House set its recommendations. Because of these two potential time lags, we define the “recommendation date” in this study to be one week *before* the actual publication date of each survey.⁴

Most of the surveys reported recommended strategic asset allocations among equities, bonds, and cash, i.e., short-term government paper. In some instances, the surveys also provided recommended allocations to other assets such as real estate, gold, and fine art. The recommended percentage allocation to these other assets averaged less than 1% throughout the sample. Therefore, we simply added this other allocation to the percentage allocation to cash.

Equity holdings in the survey are further broken down into individual country shares. The holdings for country shares are grouped as follows: North America [United States (U.S.) and others], Europe [France, Germany, the United Kingdom (U.K.) and others], and the Far East [Japan and

⁴This assumption concerning the recommendation date is not critical. Test results using the actual publication date of the surveys are nearly identical to the test results reported in the paper. In particular, there are no changes in significant results. Details available upon request.

others]. Five equity markets received, on average, 80% of the recommended equity weights. These five major equity markets are those for the U.S., Japan, the U.K., Germany, and France. Further, these five equity market categories existed throughout the sample period. The rest of the equity allocation, 20% on average, is scattered across other world equity markets.

The initial set of recommendations were by these eight Houses: Anonymous One and Anonymous Two, London Merchant Banks; Brown Brothers Harriman, Wall Street private bankers; Capital House Asset Management; Daiwa Europe Limited, a Japanese investment bank; Scudder Stevens Clark, New York investment counselors; Phillips & Drew, London brokers; and U.S. Trust. Bank Julius Baer, a Swiss investment bank, replaced U.S. Trust in March 1983. In January 1987, a ninth participant, Wardley Investor Services, the fund management arm of Hong Kong and Shanghai Banking Corporation, joined the group. The set of nine Houses remained unchanged through February 1989. Then, however, the set of Houses changed periodically through the end of the sample, October 1999. For all dates in the sample, there are at least seven Houses polled, with a maximum of ten and an average of 8.3.

Twenty four Houses participated in the surveys from 1982 to 1999. Eight of these Houses offered 10 or fewer recommendations to equities, bonds, and cash. Given the low number of recommendations, it would be difficult to evaluate the investment strategies of these Houses. Therefore, the recommendations of these eight Houses are not included in this study. Figure 1 presents a time line illustrating the survey participation dates by the sixteen remaining Houses from 1982 through 1999.⁵

2.2 Data Used to Calculate Daily Returns.

As in many other studies of international markets, this study uses the Morgan Stanley Capital International (MSCI) equity indexes for total returns (capital gains plus dividends) in U.S. dollars. The MSCI value-weighted world index serves as the proxy for the world market portfolio in tests involving strategic asset allocations. Using daily data, we compute equity index returns for the U.S.,

⁵The Houses not included in the sample, and the number of their recommendations are: Global Asset Management, 10; Deutschebank, 8; U.S. Trust, 6; Capital Management, 5; Citicorp, 3; Indocam, 2; Rabobank International, 2, and; Standard Life, 2.

Japan, the U.K., Germany, and France. In addition, because the Houses placed a nontrivial weight in the “other” equity category, we strove to construct a meaningful return series. To do so, we created an equally-weighted “rest of the world” return using MSCI indexes for the following twelve equity markets: Australia, Austria, Belgium, Canada, Denmark, Hong Kong, Italy, the Netherlands, Singapore, Spain, Sweden and Switzerland.

To calculate the daily returns on cash, we used the 3-month U.S. Treasury bill discount rate to compute a series of daily Treasury bill prices. These discount rates were obtained from Datastream©. For bond returns, however, no single daily index series spanned the sample period. Therefore, for the 1982 to 1986 sub-sample, the only data available to us was the U.S. Government Total Return Index for bonds with more than ten years to maturity, from Datastream©. However, for the rest of the sample, a world government bond price index series was created from two Datastream© sources. For the 1986 to 1991 and the 1993 to 1999 sub-samples, the daily bond return series was created using the J. P. Morgan Global Government Bond Price Index in U.S. Dollars. Because this data series was not available daily in 1992 from Datastream©, the 1992 return series was created using the J. P. Morgan Global Broad Government Total Return Index in U.S. Dollars⁶

3 Sample Statistics.

3.1 Asset Allocations Among Equity, Bonds, and Cash.

The survey questions asked during the sample period are not uniform. That is, there are some periods when the survey reported recommendations for asset allocations among equity, bonds, and cash, some periods when the survey reported recommendations for equity allocations among countries, and some periods where both were reported.

Strategic asset allocation recommendations for equity, bonds, and cash appear in Tables 1 and 2. The number of recommendations for equity, bonds, and cash ranged from 22 to 97. Eight Houses

⁶To examine some properties of this data series, we calculated the correlation coefficient between returns using the U.S. Government Total Return Index for bonds with more than 10 years to maturity and the J.P. Morgan Global Government Bond Index from 1986 to 1999. The correlation coefficient was 0.80. In addition, we ran several momentum tests using the U.S. data in place of the J. P. Morgan Indexes. There were no material changes in the results.

made at least 60 recommendations, with Daiwa Europe and UBS Phillips & Drew making the most, with 97 and 96, respectively.⁷

On average across the sample period, the Houses recommended that investors allocate 66.1 percent of their portfolio to equities, 25.4 percent to bonds, eight percent to cash, and 0.5 percent to other. One striking feature of the recommendations is the wide divergence of opinion among the Houses. Average equity holding recommendations ranged from 37.0 percent to 91.5 percent. At times, several Houses recommended that investors place 100 percent of their portfolio into equity. In every case, each House recommended that investors place a minimum of 20 percent in equity.

Table 1 also contains additional information about the number of increases recommended for equity, the size of the increases, the number of decreases recommended for equity, and the size of the decreases. Note that, on average, the Houses recommended increasing equity holdings 12.8 times and recommended decreasing equity holdings 11.3 times. Given that the average number of recommendations is 50.8, the Houses changed their strategic asset weights about half the time. The average increase in the weight across all Houses is 6.5% and the average decrease in the weight across all Houses is 6.9%. In general, the average increase and the average decrease are also symmetric by investment House.⁸

Average bond holding recommendations also appear in Table 1. These averages ranged from 1.3 percent to 50.7 percent. Throughout the sample period, Bank Julius Baer and Credit Suisse are the biggest “fans” of bonds—“perhaps partly reflecting traditional Swiss preference for fixed-interest paper” [*Financial Report*, March 17, 1983]. At no time did any House recommend that investors place more than 70 percent into bonds. During the sample period, there were six Houses that recommended, at least once, that bonds should receive a zero percent allocation.

Differences in opinion are common throughout the sample period. For example, consider the asset allocations for the two months surrounding the October 1987 market “crash.” The September 1987

⁷However, due to a time period late in the sample when strategic asset allocation recommendations were not reported, a maximum of 94 recommendations are subsequently analyzed.

⁸It is important to note that the average increases and average decreases reported in Table 1 are not percentage changes. Rather, they represent the average of the increase or decrease in the allocation percentage. For example, if a House changes its equity allocation recommendation from 50% to 60%, we report this as a 10% increase in the weight, not a 20% change in the weight.

equity recommendations ranged from 50 to 98 percent; bonds ranged from zero to 35 percent; and cash ranged from two to twelve percent. Thus, in September 1987, the Houses are still confident about the prospects for equity markets, but appear gloomy about interest rates, because most Houses recommended low allocations to bonds. In November 1987, all but one House reduced equity allocations and increased cash allocations. The exception, Scudder Stevens Clark, increased its recommended equity allocation from 97 percent to 99 percent, and decreased both bond and cash allocations. In the report that accompanies the survey, the Houses admit that “Black Monday” caught them by surprise. Further, most of the Houses did not believe that equity markets had reached a bottom, nor did they think that bonds were an attractive alternative to equity. Their opinion was that bond markets had been beset by inflationary fears, while the equity markets were restrained by fears of recession. Therefore, the Houses saw bonds as being less flexible than cash, while carrying too much risk to offset prospective gains.

Figure 2 illustrates the time variation in average asset allocations. At the beginning of the sample period, April 1982, the average equity allocation was 56.1 percent. This average increased to 65.7 percent in September 1982, and peaked at 82.7 percent in January 1984. Here, two Houses, Anonymous One and Anonymous Two, allocated their entire portfolio to equities. There was little variation in average equity allocation recommendations until 1987, when the average dropped from 78.3 percent in September to 64.9 percent in December. Subsequently, the average rose to the low-70 percent levels in late 1988. Following a drop to 65.5 percent in early 1989, the average rebounded to the mid-70 percent levels before beginning a multi-year slide, bottoming out at 45 percent in October 1992.

The average recommended bond allocation at the beginning of the sample period was 31.3 percent, but fell to 15.2 percent in April 1983, reflecting the nervousness about interest rates by the Houses. The average bond recommendation increased slightly until mid-1985, when it began a slow fall to another bottom of 14 percent in August 1987. Bond recommendations were stable in the 14 to 18 percent range until December 1990, when the average recommendation jumped to 26.3 percent. Thereafter, the average bond allocation rose to a maximum of 45.4 percent in October 1992, and then held steady at about the 40 percent level until July 1997.

The average recommended cash allocation at the beginning of the sample was 12.7 percent. It increased to 14.8 percent in July 1982, and then dropped off to 2.8 percent by January 1984. There are three spikes in the average cash allocation recommendation. In 1984, citing uncertainty about interest rates and the dollar, the Houses increased their cash allocation from 2.8 percent in January to 11.5 percent in June. Cash allocation then decreased to a low of 2.6 percent in March 1986. An increase in average cash holdings in late 1987 from 6.2 percent in September to 16.0 percent in December corresponded to the average decrease in equity holdings. The opinion of the Houses during that period was that “Cash is king” [*Financial Report*, December 10, 1987]. The third dramatic increase in recommended cash holdings occurred in September 1990, about the time of the Iraqi invasion of Kuwait. The average cash recommendation increased from 13.8 percent in July 1990 to 22.7 percent in September 1990. After the conclusion of the Gulf War, the average cash recommendation fell to 8.7 percent in April 1991.

3.2 Equity Allocations Across Countries.

Figure 3 illustrates the time variation in average equity allocations across countries. This data is summarized in Table 2. On average, all Houses recommended that investors allocate the largest percentage of their equity holdings, 40.8 percent, to the U.S. The average allocation recommended for U.S. equity during the sample period ranged from 30.9 to 48.2 percent. There was a broad range of opinion concerning U.S. equity holdings during the period. For example, UBS Phillips & Drew recommended a high of 68.0 percent and a low of 2.9 percent; Daiwa Europe had a high of 62.5 percent and a low of 7.0 percent.⁹

The next largest average recommended equity allocation, 21.6 percent, was made to Japan. No House had a higher average equity allocation to Japan than to the U.S. There was also a broad range of equity allocations to Japan. For example, Capital House had a high of 57.5 percent and a low of 7.0 percent and Daiwa Europe had a high of 45.0 percent and a low of 3.0 percent.

⁹In part of the survey, equity allocation recommendations made by Scudder Stevens and Brown Brothers Harriman are put together by their international investment departments that deal only with non-American investment. To account for this, the *Financial Report* assumed a 50 percent stake in U.S. equities for these two Houses during some of the sample.

The smallest recommended portfolio allocations in the sample were for France. To a large extent, this reflected concerns about the Mitterand government that came into power in 1981. Until April 1983, only one House, Phillips & Drew stayed in the French market after the Socialists came to power. In April 1983, three Houses, Anonymous One, Scudder Stevens, and Brown Brothers Harriman, included French equities in their portfolios. The remaining Houses did not recommend the French equity market until 1986. The 1986 French elections, which took a conservative change, as well as lower inflation in France, contributed to a more favorable investment climate.

Throughout the sample, the Houses recommended allocating an average of nearly 20 percent to countries other than the U.S., Japan, the U.K., Germany and France. These recommended allocations ranged from a low of 13.4 percent in June 1986 to a high of 27.4 percent in September 1990. Unfortunately, the breakdowns are not always made in a way to indicate which countries comprise the “other” category. Quite often in the survey, other equity allocations are made only to such broad categories as ‘Other North America’, ‘Other Asian,’ or ‘Other European.’ As a result, we assumed that this category was comprised of twelve equity markets listed in Section 2.2.

4 Momentum Trading Test Methods.

This section presents the methods employed before statistical tests were conducted to answer the main research question of the paper. Specifically, did the Houses recommend allocation changes based on past return performance of the constituent assets?

4.1 Buying Sentiment.

Our sample consists of a cross-section of recommendations made by sets of different Houses over a time-series of recommendation dates. To assess buying sentiment for each recommendation date j , we use the buying sentiment ratio introduced by Lakonishok, Shleifer, and Vishny (1992) to calculate the following proportion for equities, bonds, and cash, represented by i :

$$N_{ratio}(i) = \frac{B(i)}{Active(i)} \tag{1}$$

In equation (1), $B(i)$ represents the number of Houses recommending an increased weight in asset class i , and $Active(i)$ represents the number of Houses recommending a change in the weight for asset class i at a particular recommendation date j . Note that if there is no change in a weight, there is no entry into the calculation of the buying sentiment ratio.

We then measure the sentiment ratio for equities, bonds, and cash as follows. At each recommendation date j , we compute annualized returns for equities, bonds, and cash for the period between recommendation dates. For each recommendation date, we rank the returns for equities, bonds, and cash. The sentiment measure is then assigned to the corresponding performance rank. For each performance rank, we calculate the average value for $N_{ratio}(i)$ across all recommendation dates at which at least one investment House changed its recommended asset weight.

4.2 Allocation Changes.

To provide further evidence and to add robustness, we also investigate changes in the allocations among equities, bonds, and cash. At each recommendation date, the allocation change for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. As with the sentiment ratio, for each recommendation date, we rank annualized returns for the period between recommendation dates among equities, bonds, and cash. Then, for all cases where equities, bonds, and cash rank best, worst, or in-between, we calculate the average change in the recommended allocation.

5 Empirical Results.

5.1 Momentum Tests Concerning Strategic Asset Allocations.

We conduct statistical tests of the difference in the sentiment levels for the best and worst return performance ranks across asset allocations. Grinblatt, Titman and Wermers (1995) point out that ordinary t-statistics are technically inappropriate when portfolios have weights that change over time. To account for this statistical issue and to add robustness, we also calculate Spearman’s rank correlation statistic across performance ranks and perform Wilcoxon rank-sum tests between best

and worst rankings.

Panels A and B of Table 3 contain the results of the tests for momentum trading for strategic asset allocations. There is strong evidence that House recommendations, on average, are consistent with the use of momentum trading strategies by the Houses.

5.1.1 Equities and Cash.

When equity performance has been the best of the three asset classes, 58 percent of the Houses increase their recommended equity holding. However, when equity performance has been the worst, only 37 percent of the Houses recommend increasing the allocation to equities. These differences are significant at the five percent level using the Wilcoxon rank-sum test or a t-test. Further, using a Spearman's rank correlation statistic, the ranking of equity allocations based on past performance is also significant at the five percent level.

The changes in equity allocations are similar. When equity performance has been the best, the Houses, on average, increased their recommended equity allocations by 0.79 percent. When equity performance has been the worst, the Houses, on average, recommend a decrease of 2.29 percent. These differences are significant at the one percent level, using the Wilcoxon rank-sum test or a t-test. Further, using a Spearman's rank correlation statistic, the ranking of equity allocations based on past performance is also significant at the one percent level.

The results also suggest that the Houses used a momentum strategy for cash recommendations. When cash has been the best performing asset, on average, 63 percent of the Houses increased their recommended cash allocation. On average, they increased their recommendation by 0.73 percent in this case. However, when the return on cash has been the lowest, only 35 percent of the Houses increased their cash allocation. On average in this case, the Houses decreased their allocation to cash by 0.89 percent.

5.1.2 Bonds.

In contrast to the strong evidence for equities and cash, there is no evidence that the Houses used a momentum trading strategy for bonds. To further investigate this finding, we sub-divided

the sample into two groups as follows. At each recommendation date, we compute the average recommended bond holdings across all Houses. Then, we define “High-Bond Houses” to be those that recommend bond holdings at a level greater than the mean and define “Low-Bond Houses” to be those that recommend bond holdings at a level less than the mean. Momentum trading results for the High-Bond Houses appear in Table 4 and results for the Low-Bond Houses appear in Table 5.¹⁰

There are several notable observations concerning momentum test results for each group. First, as illustrated in Panel A of Table 4, there are no significant differences in buying sentiment for equities for the High-Bond Houses. However, as illustrated in Panel A of Table 5, there is evidence consistent with the notion that the Low-Bond Houses used momentum strategies for equities. When equities have had the best performance of the three asset classes, on average, 62 percent of the Low-Bond Houses increased their recommended equity holdings. However, when equities have had the worst performance, only 40 percent of the Low-Bond Houses increased their equity holding recommendations. Using a Wilcoxon rank-sum test, this difference is significant at the one percent level.

Second, even when conditioning on the bond allocation level, there is no evidence to support the notion that the Houses engaged in momentum trading strategies concerning their bond allocations. Third, in a manner consistent with the notion of momentum trading, both High-Bond and Low-Bond Houses adjusted their cash recommendations in response to the past ranking of cash returns. Finally, as detailed below, the buying sentiment and allocation changes differed significantly in magnitude between the High-Bond and Low-Bond Houses.

Buying Sentiment. Testing for differences between numbers reported in Panel A of Tables 4 and 5 reveals that the only significant differences between High-Bond and Low-Bond Houses occurs for their bond recommendations. When bonds had the highest return, 68 percent of the High-Bond

¹⁰Throughout the sample, the Houses tended to be either a High-Bond or Low-Bond House. Houses that predominately appear in the High-Bond group are UBS Phillips & Drew, Bank Julius Baer, Credit Suisse, and Robeco Group. The Low-Bond Houses tended to be Scudder Stevens Clark, Brown Brothers Harriman, Lehman Brothers, Nikko Securities, Credit Agricole, Anonymous 1, and Anonymous 2. The remaining five Houses appear in each of the two bond groups about half the time they make a recommendation.

Houses increased their bond allocation recommendation. However, in this case, only 43 percent of the Low-Bond Houses increased their recommended allocation to bonds. Also, when bonds had the lowest return, 57 percent of the High-Bond Houses increase their bond holding recommendations, compared to only 38 percent of the Low-Bond Houses. These differences in sentiment are both significant at the five percent level.

Allocation Changes. As shown in Panel B of Tables 4 and 5, when equities have performed the best of the three asset classes, High-Bond Houses increased their equity holding recommendation an average of 0.11 percent. However, on average, Low-Bond Houses increased their equity recommendations by 1.36 percent. A t-test for the difference in these recommendations yields a t-statistic of 2.23, significant at the 2 percent level. Similarly, when equities have performed the worst, High-Bond Houses decrease their equity recommendations by 3.90 percent, while Low-Bond Houses decrease their holdings by 1.23 percent. However, this latter difference is significant only at the 10 percent level.

A further examination of data in Panel B of Tables 4 and 5 also reveals a difference between High-Bond Houses and Low-Bond Houses concerning bond allocations. When bonds returns were the highest, the High-Bond Houses increased their recommended bond holdings by 3.19 percent. In contrast, the Low-Bond Houses decreased their recommended bond holdings by 0.94 percent in this case. Although not reported in the table, the t-statistic for this difference is -3.53, which is significant at the one percent level. Finally, when bond returns were the lowest, the High-Bond Houses increased their recommended bond holdings by 0.58 percent, while the Low-Bond Houses decreased their recommended bond holdings by 1.12 percent. The unreported t-statistic for this difference is -2.94, which is also significant at the one percent level.

5.2 Momentum Tests Concerning Tactical Equity Allocations.

We calculate both a sentiment index, $N_{ratio}(i)$, and allocation changes for equity allocations across countries in the same manner outlined above for broad asset allocations. The only difference is that we replace equity, bond and cash allocation recommendations with recommendations for equity

allocations among the United States, the United Kingdom, France, Germany, Japan and “Other” equity. Of course, this results in six, rather than three, rankings.

5.2.1 All Houses.

Test results for momentum trading behavior in country-specific, or tactical, equity allocation recommendations for all Houses appear in Panels C and D of Table 3. The evidence here suggests that the Houses did respond to past equity returns of the U.S., Japan, the U.K., and Germany. The results are strongest for the U.S. and Japan.

When the U.S. returns were the best among the six constituents, 56 percent of the Houses recommended an increase in U.S. equities. When the U.S. returns were the worst, only 31 percent of the Houses increased their recommended U.S. equity holdings. These differences are significant at the 5 percent level using either a t-test or a Wilcoxon rank-sum test. For allocation changes, the Houses increased U.S. equities by 0.73 percent when U.S. returns were best, but decreased their recommended U.S. allocation an average of 1.01 percent when U.S. returns were worst. This difference is significant at the 1 percent level using either a t-test or a Wilcoxon rank-sum test.

When Japanese equity returns were the best, 66 percent of the Houses recommended increasing Japanese holdings, but when Japanese equity returns were the worst, only 35 percent of the Houses increased their recommended Japanese equity allocation. For allocation changes, on average the Houses increased their recommended Japanese holdings by 1.03 percent when Japanese equity returns were the best, and decreased their recommended Japanese holdings by 1.53 percent when Japanese equity returns were the worst. These differences are both significant at the at the 1 percent level, using either a t-test or a Wilcoxon rank-sum test.

5.2.2 High-Bond and Low-Bond Houses.

Panels C and D of Table 4 contain test statistics for the High-Bond Houses. There is little evidence to suggest that the High-Bond Houses employed a momentum trading strategy across countries. The only significant test results occur in the Japanese equity market. Thus, it is likely that the Low-Bond Houses drove the results presented above for all Houses. Test statistics for the Low-Bond

Houses appear in Panels C and D of Table 5. The test statistics for the Low-Bond Houses are similar to the test statistics for all Houses. The exception is that no evidence is found that the Low-Bond Houses used a momentum trading strategy for the U.K.

5.2.3 Houses Grouped by Continent.

Several studies document that foreign investors pursue momentum strategies. In particular, Brennan and Cao (1997) develop and test a model to investigate the implications of an information advantage held by domestic investors over foreign investors. Brennan and Cao show that investors tend to purchase foreign assets when the return on foreign assets is high, and sell foreign assets when the return of foreign assets is low.

We examine the question of whether similar momentum trading occurred in our sample. A priori, one would expect that the strong international information network possessed by these Houses would serve to mitigate information disadvantages. Based on the location of each home office, we grouped the Houses by continent.¹¹

Table 6 presents the results of tests for momentum trading in country-specific, or tactical, equity allocation recommendations. Panel A contains the test results for the North American Houses. There is some evidence that the North American Houses employ a momentum trading strategy outside the U.S. While the only consistently significant results appear in the German equity market, the North American Houses also exhibited momentum trading in the French equity market and the Other equity market.

Panel B of Table 6 contains the momentum trading test results for the Houses based in Asia. Tests using allocation changes strongly support the notion that these Houses employed momentum trading for both the U.S. and Japanese equity markets. However, using the sentiment measure, the only support for the notion of momentum trading is found in the U.S. equity market.

Panel C of Table 6 presents the momentum trading test results for the Houses based in Europe.

¹¹The North American Houses are Brown Brothers Harriman, Lehman Brothers, Merrill Lynch and Scudder Stevens Clark. Note that all these Houses are based in the U.S. The Asian Houses are Daiwa Europe, Nikko Securities, and Wardley Investor Services. The European Houses are Anonymous One, Anonymous Two, Bank Julius Baer, Capital House, Credit Agricole, Credit Suisse, Robeco Group, and UBS Phillips & Drew.

Using either the buying sentiment measure or the allocation change, strong support for the notion of momentum trading is found for the U.S. and Japan. There is some evidence that supports the notion of momentum trading in the French and Other equity markets. Note that no House had a home office in these categories. However, there are Houses based in the U.K., and there is some evidence consistent with momentum trading in the U.K. equity market by the European Houses.

Broadly speaking, these results are consistent with Brennan and Cao (1997). That is, we find more evidence that foreign Houses employed momentum trading strategies and find sparse evidence that domestic Houses employed momentum trading strategies. Specifically, however, Brennan and Cao (1997, p. 1876) find that “while U.S. purchases of equities in developed foreign markets tend to be positively associated with the foreign market return, foreign purchases of U.S. equities show no such relation to the U.S. market return.” In our sample, there is only weak evidence that the North American (really U.S.) Houses employed momentum trading strategies in foreign markets. In addition, unlike the findings of Brennan and Cao (1997), we find that the country-specific equity allocation recommendations of Houses based in Asia and Europe changed in response to past rankings of U.S. equity returns.¹²

6 An Examination of Two Other Investment Strategies.

To this point, the only investment strategy we have investigated is momentum trading. In this section, we present a preliminary analysis to consider the possibility that the Houses used other investment strategies. Specifically, we first investigate whether there is a relation between changes in House recommendations and changes in Dividend Yields or changes in Price-Earnings Ratios. Then, we examine the relationship between changes in House recommendations and changes in macroeconomic variables.

¹²Brennan and Cao state that their finding is consistent with the notion that foreign investors are as well informed about U.S. markets as are U.S. investors, but U.S. investors are less well informed about foreign markets than the residents of those countries. Our results suggest the opposite.

6.1 Tests for a Value-Investing Strategy.

The strategy for the Houses might be a timing strategy, calling for an increased exposure to equities after equity prices have depreciated, as well as a decreased exposure to equities after equity prices have appreciated. This contrarian strategy is similar to the fundamentalist investment strategies of Graham and Dodd (1934). Graham and Dodd forward two earnings-related investment strategies: buying equities with low Price-Earnings Ratios, and buying equities with high Dividend Yields.¹³

6.1.1 Strategic Asset Allocation to Equity.

For tests concerning strategic equity recommendations, the one-month *changes* in the Price-Earnings Ratio and the Dividend Yield of the MSCI World Equity Index are used to divide the sample period into two different states. High (Low) states are defined as those where the state variable is above (below) the sample period median. Table 7 contains test results of the value-investing approach.

Price-Earnings Ratio. When the change in the Price-Earnings Ratio was less than the median change, the Houses, on average, decreased their recommended equity allocation by 0.60 percent. When the change in the Price-Earnings Ratio was greater than the median change, the Houses, on average, increased their recommended equity holdings by 0.78 percent. This difference is significant at the 1 percent level using a t-test or a Wilcoxon rank-sum test.¹⁴

These recommended changes, however, are not consistent with a value-investing strategy for the following reason. Because earnings are reported quarterly, decreases in monthly prices result in one-month decreases in the Price-Earnings Ratio. Large one-month price decreases result in large one-month decreases in the Price-Earnings Ratio. The large decreases in the Price-Earnings Ratio

¹³Testing equity markets in Japan, the U.K., Germany, and France, Hawawini and Keim (2000) find that the magnitude of the difference in returns between portfolios with low Price-Earnings Ratios and portfolios with high Price-Earnings Ratios is comparable to the difference between such portfolios in the U.S. There is some evidence that Dividend Yields predict returns on national market indexes [e.g., Poterba and Summers (1988) and Cutler, Poterba and Summers (1989)]. Keppler (1991) documents the potential benefits of investing where Dividend Yields are above average.

¹⁴We repeat this analysis, as well as the analysis for Dividend Yield, using levels of Price-Earnings Ratios and Dividend Yields. There is no statistically significant evidence that the Houses adjusted their recommendations in response to the most recent levels of the Price-Earnings Ratio or Dividend Yield. These results are available from the authors upon request.

would appear in the Low state. Hence, observing decreased recommended equity allocations in the Low state is consistent with the Houses using a momentum investing strategy. That is, decreasing equity allocations after a decrease in price. A parallel argument can be constructed for the High state.

High-Bond Houses (Table 7, Panel B) and Low-Bond (Table 7, Panel C) Houses, however, exhibit asymmetric responses to changes in the Price-Earnings Ratio. High-Bond Houses, on average, decreased equity allocations in the Low state. By contrast, Low-Bond Houses, on average, increased their equity recommendation in the High state. Both responses, however, are again consistent with the Houses using a momentum investing strategy, not a value-investing strategy.

Dividend Yield. The results using one-month changes in the Dividend Yield are similar to the results above. The Houses, on average, recommended an increase of 0.70 percent in the Low state, and, on average, recommended a decrease in equity allocations of 0.51 percent in the High state. Given the relative stability of cash dividends over time, one-month changes in the Dividend Yield are most likely driven by changes in prices. If so, these results are once again consistent with the Houses using a momentum trading strategy. For example, consider the Low state. With stable dividends, a price increase decreases the Dividend Yield. Large one-month price increases would result in large one-month decreases in the Dividend Yield, placing these instances in the Low state. The Houses recommended increasing equity holdings in the Low state, i.e., after large one-month price increases.

6.1.2 Tactical, or Country-Specific, Equity Allocations.

For tests concerning tactical equity recommendations, we compute the one-month *changes* in the Price-Earnings Ratio and the Dividend Yield of the MSCI Indexes for the U.S., Japan, the U.K., Germany, and France. At each recommendation date j , we rank these one-month changes for the five countries, from highest to lowest. Similar to the method employed for momentum tests, we assign the buying sentiment measure for each recommendation date to the corresponding rank. Then, we calculate the average buying sentiment. For allocation changes, we assign the country equity

allocation change for each house to the appropriate ranking, and calculate the average change in recommended equity allocations.

Table 8 contains test results of the value-investing approach for tactical equity allocations. There is little evidence that the Houses used a value-investing strategy for country-specific equity allocations. The only three significant results are for equity allocation changes in response the change in the U.S. Dividend Yield (Panel A), and the changes in Price-Earnings Ratio and Dividend Yield for Japan (Panel B).

6.2 Macroeconomic Factor Approach.

We also examine the relation between growth rates in macroeconomic variables and changes in equity recommendations and buying sentiment. For our tests we draw on the existing, but often contradictory, evidence that market returns respond to macro economic developments.¹⁵

6.2.1 Strategic Asset Allocation to Equity.

We use similar statistical tests to those for the Price-Earnings Ratio and the Dividend Yield. For tests of equity recommendations, we use G-7 Inflation, the growth in G-7 Industrial Production, and the growth in the U.S. Dollar Trade-Weighted exchange rate between recommendation dates. We divide the sample period into two different states. High (Low) states are defined as those where the state variable is above (below) the sample period median. The tests for equity recommendations are reported in Table 7.

Inflation is the only macroeconomic factor with a statistically significant relation to changes in the recommended equity allocations and the buying sentiment measure, $N_{ratio}(i)$. As seen in Panel A, all Houses, on average, recommended increasing equity holdings (0.54 percent) in the Low state,

¹⁵Researchers have documented a significant effect of inflation on equity returns [Bodie (1976), Fama (1981), Geske and Roll (1983), Pearce (1985), Pearce and Roley (1983), and Flannery and Protopapadakis (2001)]. However, the evidence for other macroeconomic variables is less compelling. Chen, Roll and Ross (1986) conclude that Industrial Production is a strong candidate for a risk factor and find weaker evidence to support inflation as a risk factor. However, Cutler, Poterba and Summers (1989) find that industrial production growth is not significantly positively correlated with real stock returns in the 1946-1985 period, which roughly corresponds to the sample period in Chen, Roll, and Ross (1986). Flannery and Protopapadakis (2001) find that real gross national product and industrial production are not priced risk factors. Lamont (2000) concludes that a portfolio constructed to “track” the growth rate of industrial production earns abnormal positive returns, while the portfolio that tracks the consumer price index does not.

and recommended a decrease in equity holdings (-0.53 percent) in the High state. The sentiment measure shows that 60 percent of the Houses recommended an increase in equity holdings in the Low state, compared to 44 percent in the High state. These differences are significant at the 5 percent level.

Test results for the High-Bond Houses (Panel B) and the Low-Bond Houses (Panel C) show that only the Low-Bond Houses respond to inflation. In the Low state, the Low-Bond Houses recommended an average increase of 1.6 percent in equity allocations. In the High state, the Low-Bond Houses recommended an average decrease in their recommended equity allocation of 0.40 percent. These differences are significant at the one percent level. For the buying sentiment measure, in the Low state, 68 percent of the Low-Bond Houses, on average, recommended an increase in equity allocations. In the High state, only 41 percent of the Low-Bond Houses, on average, recommended an increase their recommended equity allocations. These differences are also significant at the one percent level.

6.2.2 Tactical, or Country-Specific, Equity Allocations.

We repeated the tests for a macroeconomic variable investment approach using changes in country-specific equity recommendations. In these tests, we used country-specific Inflation, growth in Industrial Production, and growth in the the U.S. Dollar Exchange Rate. For the U.S. currency, we use the U.S. Dollar Trade-Weighted Exchange Rate. As reported in Table 9, there is generally no evidence that the Houses used a macroeconomic investing strategy for country-specific equity allocations.

In summary, we find little evidence that the Houses adjusted their equity recommendations in response to past movements in macroeconomic factors. This lack of results may reflect the Houses' belief that, as Fama (1990) argues, equity price changes should predict future macro conditions, because equity prices reflect expected future cash flows.

7 Conclusion.

A significant contribution of this study is that we are able to test for the existence of momentum trading strategies in a global arena by using strategic and tactical asset allocation recommendations revealed in a time-series of surveys of a panel of investment houses (the “Houses.”)

In this study, we examine the impetus for recommended changes in portfolio allocations. Specifically, we explore whether past market performance influences modifications to weight recommendations across asset broad asset classes of stocks, bonds, and cash as well as equity allocations across six country categories.

We find strong evidence that the Houses, as a group, adjusted their strategic allocations across equity and cash in response to recent returns in these broad asset classes. We also find strong evidence that the Houses, as a group, change their tactical allocations across country-specific equity weights in response to relative return rankings across countries. These findings are consistent with the notion that the Houses engaged in momentum trading. In addition, the nature of our findings concerning tests of two alternative investment strategies buttresses these findings.

Our results are significant to future research on portfolio recommendations across market sectors, mutual fund classifications, or international markets in two areas. First, Badrinath and Wahal (2000) examine momentum trading by institutions and report results that buttress the findings of Lakonishok, Shleifer and Vishny (1992). That is, these researchers find evidence that the varied investment styles and strategies of institutional investors tend to offset one another. In our sample of sixteen Houses, however, we do not find evidence that varied investment styles and strategies offset one another for strategic or tactical asset recommendations.

Second, Brennan and Cao (1997) conclude that purchases of equities in developed foreign markets by U.S.-based investors tend to be positively associated with the foreign market return, but that foreign purchases of U.S. equities are unrelated to returns in the U.S. market. By contrast, we find no evidence that U.S.-based Houses employed momentum trading strategies in foreign equity markets, but we do find evidence consistent with the notion that Asian and European-based Houses employed momentum trading strategies in the U.S. equity market.

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Figure 1. Survey Participation Dates by Investment House, 24 Participation Dates Minimum

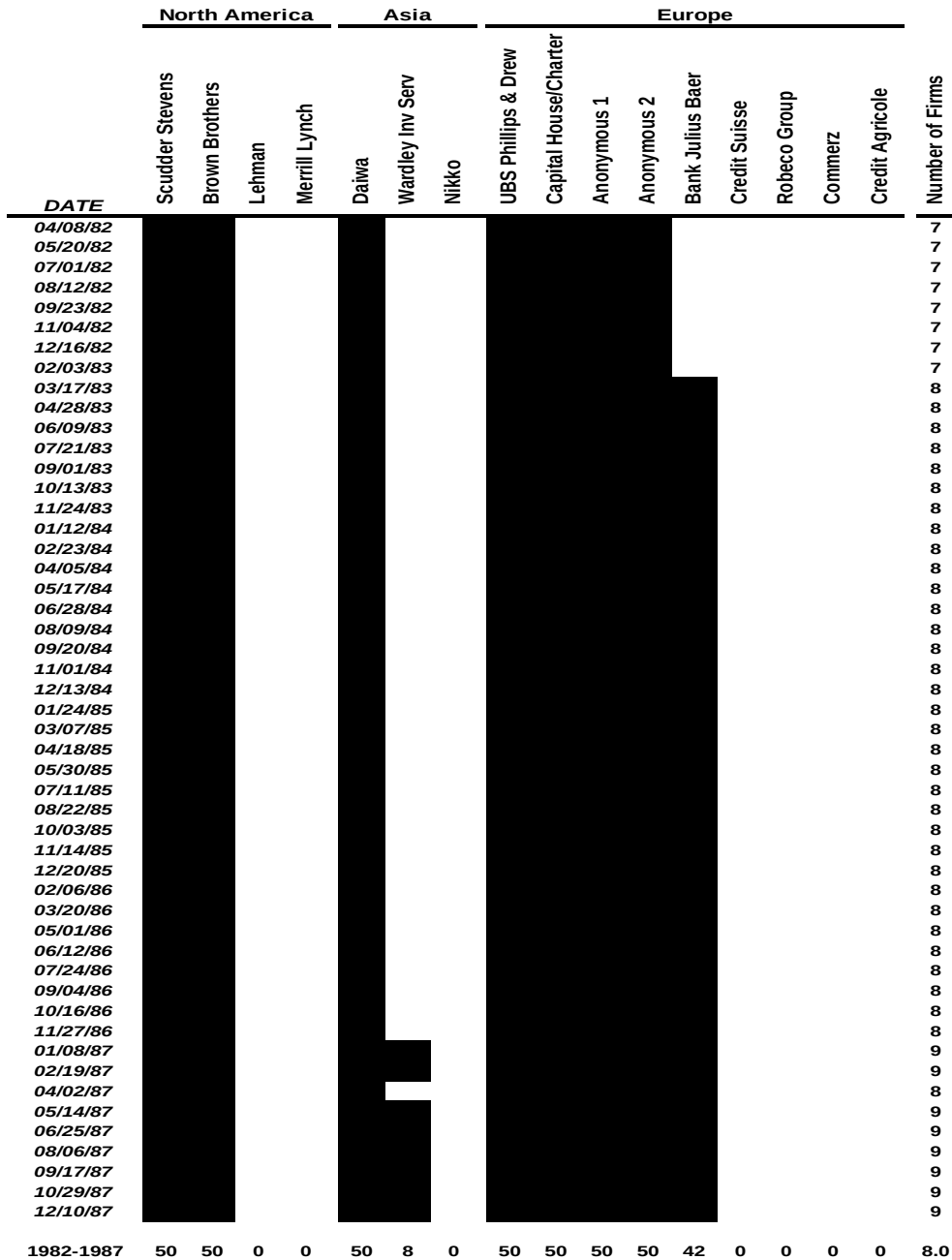


Figure 1, Cont.

	North America				Asia			Europe									
	Scudder Stevens	Brown Brothers	Lehman	Merrill Lynch	Daiwa	Wardley Inv Serv	Nikko	UBS Phillips & Drew	Capital House/Charter	Anonymous 1	Anonymous 2	Bank Julius Baer	Credit Suisse	Robeco Group	Commerz	Credit Agricole	Number of Firms
DATE																	
01/21/88																	9
03/03/88																	9
04/14/88																	9
05/26/88																	9
07/07/88																	9
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10/10/98																	7
01/16/99																	7
04/24/99																	6
07/17/99																	7
10/09/99																	6
1988-1999	22	22	28	26	49	21	27	51	22	10	10	42	32	31	35	24	8.5
1982-1987	50	50	0	0	50	8	0	50	50	50	50	42	0	0	0	0	8.0
Grand Total	72	72	28	26	99	29	27	101	72	60	60	84	32	31	35	24	8.3

Figure 2.
Average Recommended Asset Allocations Among
Equities, Bonds, and Cash.

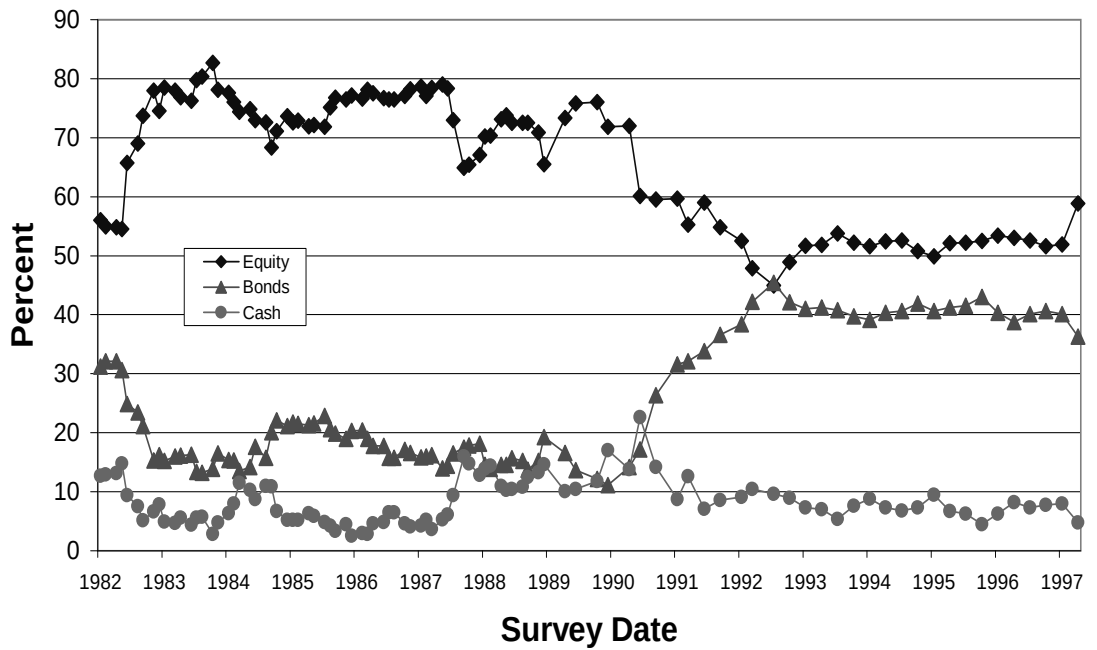


Figure 3.

Average Recommended Equity Allocation Among the U.S., Japan, the U.K., Germany, France, and All Other Countries

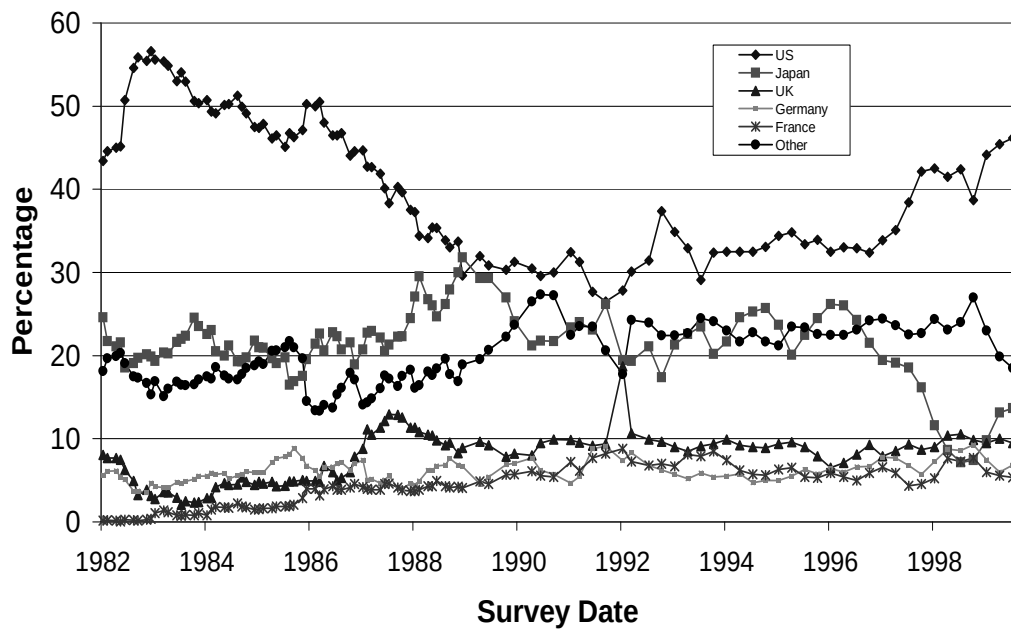


Table 1.

Recommended Percentage Allocations to Equities, Bonds, Cash, and Other Assets, 1982–1999, Listed by Number of Recommendations. This table includes the number of times each House provided a recommendation in this part of the survey and their average, high, and low percentage allocation recommendation. For equities, the table also includes the number of times the House increased and decreased their recommended equity allocation and the average size of the increase and decrease.

House	Recom- mendations	Percent Equity:			Num. of Inc.	Avg. % Increase ¹	Num. of Dec.	Avg. % Decrease ¹
		Avg.	High	Low				
Daiwa Europe	97	67.4	90	20	17	10.6	10	14.5
UBS Philips & Drew	96	53.2	70	35	13	7.7	18	6.9
Bank Julius Baer	79	41.2	53	30	13	5.1	11	6.6
Brown Brothers Harriman	72	80.4	95	45	22	5.0	12	9.6
Capital House	72	65.6	90	20	17	10.1	19	7.2
Scudder Stevens Clark	72	85.6	100	58.7	31	4.2	33	3.2
Anonymous One	60	84.8	100	60	16	6.1	10	8.6
Anonymous Two	60	91.5	100	60	13	4.1	10	8.3
Commerz International	30	53.0	63	35.5	7	7.5	9	5.8
Wardley Investor Services	29	77.7	100	50	11	9.2	10	10.2
Credit Suisse	27	37.0	50	25	7	6.7	8	3.0
Robeco Group	26	49.9	60	40	8	4.4	7	5.1
Lehman Brothers	25	59.6	86	45	7	6.9	4	7.0
Merrill Lynch	24	50.0	60	40	3	8.3	6	7.5
Credit Agricole	22	57.3	70	40	5	6.8	6	7.0
Nikko Securities	22	66.4	80	55	7	5.7	4	5.0
Average for all firms: ²	50.8	66.1	100	20	12.8	6.5	11.3	6.9

¹The average percent increase and average percent decrease is not a percentage change. Rather, it represents the average of the increase or decrease of the allocation percentage. For example, if a House changes the equity allocation recommendation from 50% to 60%, the data would reflect a 10% increase, not a 20% increase.

²The ‘High’ and ‘Low’ entries are the highest and lowest values reported over all firms across all forecast dates.

Table 1, Cont.

Recommended Percentage Allocations to Bonds, Cash, and Other Assets, 1982–1999. The data in this part of the table includes the average, high, and low percentage allocations, listed by number of recommendations.

House	Bonds:			Cash:			Other: ¹		
	Avg.	High	Low	Avg.	High	Low	Avg.	High	Low
Daiwa Europe	26.9	70	10	5.7	45	0	0.0	–	–
UBS Philips & Drew	40.3	57	23.5	5.8	20	0	0.3	5	0
Bank Julius Baer	46.9	65	35	8.9	20	0	3.0	10	0
Brown Brothers Harriman	6.5	48	0	13.2	30	5	0.0	–	–
Capital House	20.2	50	0	12.4	50	0	1.8	10	0
Scudder Stevens Clark	8.1	20.9	0	6.3	20.4	0	0.0	–	–
Anonymous One	7.5	30	0	7.8	40	0	0.0	–	–
Anonymous Two	1.3	30	0	7.1	19	0	0.3	3	0
Commerz International	42.6	50	33	4.4	24.5	0	0.0	–	–
Wardley Investor Services	13.8	45	0	8.5	45	0	0.0	–	–
Credit Suisse	50.7	64	40	12.3	27	4	0.0	–	–
Robeco Group	47.4	60	40	2.7	10	0	0.0	–	–
Lehman Brothers	31.2	45	11	8.8	20	0	0.4	5	0
Merrill Lynch	41.9	55	30	8.1	15	0	0.0	–	–
Credit Agricole	34.9	50	25	7.8	20	0	0.0	–	–
Nikko Securities	26.8	35	15	6.8	15	0	0.0	–	–
Average for all firms: ²	25.4	70	0	8.0	50	0	0.5	10	0

¹The ‘Other’ category primarily represents gold. However, there are a few recommendations that include categories such as real estate and fine art.

²The ‘High’ and ‘Low’ entries are the highest and lowest values reported over all firms across all forecast dates.

Table 2.

Recommended Percentage Equity Allocations to the United States, Japan, United Kingdom, Germany, France, and All Other Countries, 1982–1999, Listed by Number of Recommendations. This table includes the number of times each House provided a recommendation in the part of the survey dealing with country equity, their average, high, and low percentage allocation recommendation.

House	N	United States			Japan			United Kingdom		
		Ave	High	Low	Ave	High	Low	Ave	High	Low
UBS Philips & Drew	101	39.6	68	2.9	18.4	30	5	9.5	71	2
Daiwa Europe	99	40.1	62.5	7	30.3	45	3	11.6	35	5
Bank Julius Baer	84	41.4	67	20	16.8	34	0	5.6	30	0
Brown Brothers Harriman	72	48.2	50	27	18.6	25.5	12	7.1	16	2
Capital House	72	39.6	70	20	26.7	57.5	7	5.4	17	0
Scudder Stevens Clark	72	46.8	50	30	12.8	20	8.9	6.1	12.8	0
Anonymous One	60	45.5	75	24	24.8	38	10	5.3	12.5	0
Anonymous Two	60	42.4	54	25	25.0	40	15	5.1	15	0
Commerz International	35	38.4	55	23	22.4	45	6	7.2	15	3
Credit Suisse	32	35.1	50	23	18.0	30	0	8.1	13	0
Robeco Group	31	36.4	53	20	20.4	36	4	9.4	16	2
Wardley Investor Services	29	30.9	50	21	23.0	42	12	10.3	18	4
Lehman Brothers	28	41.1	55	20	12.7	26	5	9.8	17	4
Nikko Securities	27	36.2	44	23	28.9	36	6	9.8	17	7
Merrill Lynch	26	35.8	46.5	24	24.5	33.2	16.8	9.1	14	5
Credit Agricole	24	33.1	40	26	21.1	29	11	7.5	15	3
Average for all firms: ¹	53.3	40.8	75	2.9	21.6	57.5	0	7.7	71	0

¹The ‘High’ and ‘Low’ entries are the highest and lowest values reported over all firms across all forecast dates.

Table 2, Cont.

House	N	Germany			France			All Other Countries		
		Ave	High	Low	Ave	High	Low	Ave	High	Low
UBS Philips & Drew	101	6.1	20	1	6.6	15	0.5	19.9	36	8
Daiwa Europe	99	4.4	23	0	3.0	23	0	10.5	38	0
Bank Julius Baer	84	10.0	20	4	3.6	12	0	22.6	38	13
Brown Brothers Harriman	72	6.0	11	3.5	2.4	8	0	17.8	25	12
Capital House	72	8.4	25	2	2.1	13	0	17.7	40	3.5
Scudder Stevens Clark	72	4.0	8.7	1.5	3.5	7	0	26.7	38.2	20
Anonymous One	60	4.2	11	0	3.0	7.5	0	17.2	26	6
Anonymous Two	60	4.8	11	2	3.3	6	0	19.5	30	10
Commerz International	35	6.1	12	0	3.9	16	0	21.9	34	0
Credit Suisse	32	7.8	15	4	5.5	12	0	25.6	36	18
Robeco Group	31	5.4	10	2	5.6	9	3	22.8	29	15
Wardley Investor Services	29	4.3	12	0	5.8	10	3	25.7	42	15
Lehman Brothers	28	4.3	8	0	5.0	12	1	27.2	42	19
Nikko Securities	27	8.5	29	3	5.2	20	2	11.4	16	4
Merrill Lynch	26	3.6	6.3	2	4.0	6	3	22.9	31	15
Credit Agricole	24	9.0	15	7	11.1	20	6	18.2	27	10
Average for all firms: ¹	53.3	6.1	29	0	4.1	23	0	19.7	42	0

¹The 'High' and 'Low' entries are the highest and lowest values reported over all firms across all forecast dates.

Table 3.

Momentum Trading Test Results Using All Houses. For each recommendation date (j), we compute two measures. First, a Buying Sentiment Measure, $N_{ratio}(i) = \#buys(i)/\#active(i)$, is computed for each asset (i), stocks, bonds, cash, and country equities. The number of Houses recommending an increase in the holding of asset (i) is given by $\#buys(i)$, while $\#active(i)$ is the number of Houses recommending a change in the holding of asset (i). Second, the allocation change for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. Then, at each recommendation date (j), the returns for each strategic or tactical asset since the previous recommendation date are ranked. We report the average $N_{ratio}(i)$ across all recommendation dates in each performance rank cell. In the performance rank cells for the allocation changes, we report the grand average across all Houses and all recommendation dates. For each asset (i) we report a Spearman's rank correlation test statistic across all performance ranks; a t-statistic for the difference in means between the best and worst values, and; a Wilcoxon Rank-Sum Test statistic between the best and worst values. One asterisk (*) or two asterisks (**) denotes significance at the 0.05 or 0.01 level, respectively.

	Performance Rank: 1 = Best; 3 = Worst			Spearman's Rank Corr.	T-Test for Difference	Wilcoxon Rank-Sum			
	1	2	3						
<i>Panel A. Buying Sentiment, $N_{ratio}(i)$</i>									
Equities	0.58	0.53	0.37	-0.22*	2.58*	2.45*			
Bonds	0.55	0.43	0.46	-0.08	0.87	1.15			
Cash	0.63	0.54	0.35	-0.37**	3.55**	3.23**			
<i>Panel B. Allocation Changes</i>									
Equities	0.79	0.05	-2.29	-0.13**	3.64**	3.90**			
Bonds	0.79	-0.15	-0.40	-0.06	1.80	1.90			
Cash	0.73	0.53	-0.89	-0.15**	3.62**	3.89**			
	Performance Rank: 1 = Best; 6 = Worst						Spearman's Rank Corr.	T-Test for Difference	Wilcoxon Rank-Sum
	1	2	3	4	5	6			
<i>Panel C. Buying Sentiment, $N_{ratio}(i)$</i>									
United States	0.56	0.63	0.49	0.48	0.52	0.31	-0.26**	2.47*	2.31*
Japan	0.66	0.55	0.40	0.52	0.46	0.35	-0.37**	4.21**	3.68**
United Kingdom	0.61	0.50	0.49	0.51	0.46	0.41	-0.18	2.26*	2.17*
Germany	0.62	0.56	0.40	0.52	0.47	0.46	-0.24*	2.24*	2.28*
France	0.56	0.77	0.44	0.48	0.44	0.44	-0.21*	0.94	0.88
Other	0.57	0.50	0.52	0.55	0.47	0.38	-0.11	1.59	1.72
<i>Panel D. Allocation Changes</i>									
United States	0.73	0.78	-0.62	-0.63	-0.03	-1.01	-0.13**	2.77**	2.91**
Japan	1.03	0.43	-0.22	0.31	-0.03	-1.55	-0.22**	5.87**	6.14**
United Kingdom	-0.15	-0.04	0.14	0.10	0.48	-0.41	-0.10**	0.18	2.87**
Germany	0.39	-0.09	-0.06	0.12	-0.03	-0.03	-0.08*	1.55	2.38*
France	-0.05	0.71	-0.09	-0.08	-0.12	-0.04	-0.09**	-0.23	0.61
Other	0.81	-0.23	0.10	0.36	-0.34	-0.49	-0.06	1.81	1.43

Table 4.

Momentum Trading Test Results Using High-Bond Houses. For each recommendation date (j), we compute the average bond holding recommendation across Houses. Houses recommending a bond holding greater than the mean are classified as High-Bond Houses. As before, for each recommendation date (j), we compute two measures. First, a Buying Sentiment Measure, $N_{ratio}(i) = \#buys(i)/\#active(i)$, is computed for each asset (i), stocks, bonds, cash, and country equities. The number of Houses recommending an increase in the holding of asset (i) is given by $\#buys(i)$, while $\#active(i)$ is the number of Houses recommending a change in the holding of asset (i). Second, the allocation change for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. Then, at each recommendation date (j), the returns for each strategic or tactical asset since the previous recommendation date are ranked. We report the average $N_{ratio}(i)$ across all recommendation dates in each performance rank cell. In the performance rank cells for the allocation changes, we report the grand average across all Houses and all recommendation dates. For each asset (i) we report a Spearman's rank correlation test statistic across all performance ranks; a t-statistic for the difference in means between the best and worst values, and; a Wilcoxon Rank-Sum Test statistic between the best and worst values. One asterisk (*) or two asterisks (**) denotes significance at the 0.05 or 0.01 level, respectively.

	Performance Rank: 1 = Best; 3 = Worst						Spearman's	T-Test for	Wilcoxon
	1	2	3				Rank Corr.	Difference	Rank-Sum
<i>Panel A. Buying Sentiment, $N_{ratio}(i)$</i>									
Equities	0.50	0.43	0.32				-0.15	1.28	1.37
Bonds	0.68	0.43	0.57				-0.04	0.85	0.81
Cash	0.71	0.53	0.35				-0.32**	2.69*	-2.47*
<i>Panel B. Allocation Changes</i>									
Equities	0.11	-0.87	-3.90				-0.12*	2.52**	2.46**
Bonds	3.19	0.23	0.58				-0.06	2.19*	1.59
Cash	0.70	0.43	-1.19				-0.16**	3.07**	2.80**
	Performance Rank: 1 = Best; 6 = Worst						Spearman's	T-Test for	Wilcoxon
	1	2	3	4	5	6	Rank Corr.	Difference	Rank-Sum
<i>Panel C. Buying Sentiment, $N_{ratio}(i)$</i>									
United States	0.53	0.54	0.37	0.54	0.55	0.37	-0.08	1.05	1.05
Japan	0.66	0.61	0.30	0.36	0.39	0.39	-0.27**	2.24*	2.13*
United Kingdom	0.48	0.35	0.65	0.56	0.37	0.34	-0.10	0.97	0.85
Germany	0.62	0.48	0.52	0.57	0.54	0.83	-0.05	-0.80	-0.08
France	0.68	0.95	0.41	0.41	0.29	0.46	-0.34**	1.15	1.16
Other	0.58	0.36	0.57	0.57	0.67	0.17	0.10	2.18	1.76
<i>Panel D. Allocation Changes</i>									
United States	1.04	-0.15	-1.14	0.12	0.40	-1.54	-0.07	2.26*	1.78
Japan	0.55	1.10	-0.39	0.15	0.02	-1.32	-0.18**	2.46**	2.85**
United Kingdom	-0.74	-0.31	0.54	0.44	-0.30	-0.38	-0.10	-0.35	1.75
Germany	0.21	-0.04	-0.12	0.67	-0.12	-0.11	-0.01	0.30	0.26
France	0.08	0.69	-0.14	0.05	-0.20	0.28	-0.17**	-0.62	1.38
Other	0.05	-0.62	0.41	0.43	-0.06	-0.67	-0.02	0.75	0.85

Table 5.

Momentum Trading Test Results Using Low-Bond Houses. For each recommendation date (j), we compute the average bond holding recommendation across Houses. Houses recommending a bond holding less than the mean are classified as Low-Bond Houses. As before, for each recommendation date (j), we compute two measures. First, a Buying Sentiment Measure, $N_{ratio}(i) = \#buys(i)/\#active(i)$, is computed for each asset (i), stocks, bonds, cash, and country equities. The number of Houses recommending an increase in the holding of asset (i) is given by $\#buys(i)$, while $\#active(i)$ is the number of Houses recommending a change in the holding of asset (i). Second, the allocation change for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. Then, at each recommendation date (j), the returns for each strategic or tactical asset since the previous recommendation date are ranked. We report the average $N_{ratio}(i)$ across all recommendation dates in each performance rank cell. In the performance rank cells for the allocation changes, we report the grand average across all Houses and all recommendation dates. For each asset (i) we report a Spearman's rank correlation test statistic across all performance ranks; a t-statistic for the difference in means between the best and worst values, and; a Wilcoxon Rank-Sum Test statistic between the best and worst values. One asterisk (*) or two asterisks (**) denotes significance at the 0.05 or 0.01 level, respectively.

	Performance Rank: 1 = Best; 3 = Worst						Spearman's Rank Corr.	T-Test for Difference	Wilcoxon Rank-Sum
	1	2	3						
<i>Panel A. Buying Sentiment, $N_{ratio}(i)$</i>									
Equities	0.62	0.53	0.40				-0.26**	2.48*	2.53**
Bonds	0.43	0.39	0.38				-0.03	0.42	0.34
Cash	0.60	0.53	0.35				-0.28**	2.45*	2.45*
<i>Panel B. Allocation Changes</i>									
Equities	1.36	0.69	-1.23				-0.15**	2.82**	3.14**
Bonds	-0.94	-0.47	-1.12				-0.07	0.26	1.32
Cash	0.75	0.61	-0.64				-0.14**	2.25*	2.70**
	Performance Rank: 1 = Best; 6 = Worst						Spearman's Rank Corr.	T-Test for Difference	Wilcoxon Rank-Sum
	1	2	3	4	5	6			
<i>Panel C. Buying Sentiment, $N_{ratio}(i)$</i>									
United States	0.66	0.60	0.58	0.23	0.41	0.26	-0.33**	2.69**	2.37*
Japan	0.67	0.51	0.42	0.58	0.52	0.39	-0.27**	3.09**	2.91**
United Kingdom	0.67	0.65	0.47	0.33	0.55	0.45	-0.23**	1.90	1.91
Germany	0.64	0.59	0.39	0.50	0.43	0.41	-0.24*	2.02*	1.93
France	0.45	0.68	0.52	0.63	0.50	0.45	-0.06	-0.02	0.08
Other	0.57	0.61	0.46	0.53	0.33	0.46	-0.20	0.99	0.97
<i>Panel D. Allocation Changes</i>									
United States	0.53	0.88	0.08	-1.25	-0.45	-0.58	-0.15**	1.41	2.34*
Japan	1.36	-0.03	0.15	0.17	-0.20	-1.40	-0.20**	4.65**	4.37**
United Kingdom	0.21	0.15	-0.04	-0.36	1.09	-0.27	-0.09	1.80	1.88
Germany	0.33	-0.19	0.05	-0.06	-0.03	-0.16	-0.15**	1.43	3.03**
France	-0.09	0.42	-0.06	0.04	-0.13	-0.21	-0.03	0.40	-0.23
Other	1.30	0.04	0.10	0.48	-0.57	-0.12	-0.13**	1.64	1.16

Table 6.

Momentum Trading Results Using Houses Grouped by Continent. As before, for each recommendation date (j), we compute two measures. First, a Buying Sentiment Measure, $N_{ratio}(i) = \#buys(i)/\#active(i)$, is computed for each asset (i), stocks, bonds, cash, and country equities. The number of Houses recommending an increase in the holding of asset (i) is given by $\#buys(i)$, while $\#active(i)$ is the number of Houses recommending a change in the holding of asset (i). Second, the allocation change for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. Then, at each recommendation date (j), the returns for each strategic or tactical asset since the previous recommendation date are ranked. We report the average $N_{ratio}(i)$ across all recommendation dates in each performance rank cell. In the performance rank cells for the allocation changes, we report the grand average across all Houses and all recommendation dates. For each asset (i) we report a Spearman's rank correlation test statistic across all performance ranks; a t-statistic for the difference in means between the best and worst values, and; a Wilcoxon Rank-Sum Test statistic between the best and worst values. One asterisk (*) or two asterisks (**) denotes significance at the 0.05 or 0.01 level, respectively.

	Performance Rank: 1 = Best; 6 = Worst						Spearman's	T-Test for	Wilcoxon
	1	2	3	4	5	6	Rank Corr.	Difference	Rank-Sum
<i>Panel A. Houses Based in North America.</i>									
<i>Buying Sentiment, $N_{ratio}(i)$</i>									
United States	0.21	0.71	0.88	0.40	0.60	0.00	0.05	1.44	0.80
Japan	0.63	0.56	0.50	0.58	0.44	0.41	-0.20	1.79	1.78
United Kingdom	0.77	0.58	0.63	0.35	0.35	0.59	-0.18	1.32	1.13
Germany	0.73	0.54	0.25	0.56	0.45	0.39	-0.21	2.29*	2.10*
France	0.18	0.65	0.50	0.46	0.63	0.56	0.19	-1.75	-1.70
Other	0.83	0.70	0.44	0.55	0.39	0.50	-0.19	1.08	1.16
<i>Allocation Changes</i>									
United States	-1.02	1.09	0.35	-0.77	0.41	-1.29	0.01	0.25	-0.82
Japan	0.59	0.14	-1.18	0.49	0.02	-0.69	-0.13	1.68	1.82
United Kingdom	0.30	0.40	0.01	-0.02	-0.15	-0.15	0.06	1.48	1.11
Germany	0.35	-0.01	-0.12	0.18	-0.02	-0.25	-0.18**	2.01*	2.38**
France	-0.32	0.12	0.04	-0.01	0.22	0.15	0.12	-1.97*	-2.06*
Other	2.40	0.80	-0.15	0.18	-0.75	-0.03	-0.23**	2.06*	1.87

Table 6, Cont.

	Performance Rank: 1 = Best; 6 = Worst						Spearman's Rank Corr.	T-Test for Difference	Wilcoxon Rank-Sum
	1	2	3	4	5	6			
<i>Panel B. Houses Based in Asia.</i>									
<i>Buying Sentiment, $N_{ratio}(i)$</i>									
United States	0.75	0.55	0.22	0.29	0.33	0.38	-0.31*	2.37**	2.08*
Japan	0.75	0.50	0.67	0.50	0.56	0.75	-0.04	0.00	0.00
United Kingdom	0.41	0.64	0.61	0.38	0.33	0.43	-0.08	-0.09	-0.10
Germany	0.69	0.65	1.00	0.38	0.86	0.30	-0.22	2.40*	2.08*
France	0.58	0.86	0.50	0.31	0.33	0.56	-0.20	0.08	0.07
Other	0.50	0.63	0.17	0.39	0.50	0.28	-0.18	1.34	1.31
<i>Allocation Changes</i>									
United States	4.39	0.11	-1.65	-1.46	-0.77	-0.20	-0.26**	2.39**	2.72**
Japan	1.75	0.25	-2.11	-1.00	0.17	-1.81	-0.22**	3.06**	2.75**
United Kingdom	0.13	-0.95	0.35	0.33	-0.07	-0.81	-0.10	1.45	1.39
Germany	1.02	0.24	0.56	-0.11	0.62	-0.76	-0.15	1.90	1.80
France	-0.01	1.61	0.20	-0.31	-0.38	0.07	-0.14	-0.21	0.10
Other	1.00	-1.33	0.67	0.03	-0.10	0.18	-0.07	0.61	0.06
<i>Panel C. Houses Based in Europe.</i>									
<i>Buying Sentiment, $N_{ratio}(i)$</i>									
United States	0.54	0.60	0.48	0.50	0.56	0.30	-0.19*	2.20*	2.20*
Japan	0.73	0.58	0.47	0.53	0.51	0.32	-0.38**	4.19**	3.66**
United Kingdom	0.60	0.48	0.44	0.55	0.51	0.36	-0.16	1.96*	1.87
Germany	0.67	0.54	0.43	0.47	0.45	0.55	-0.13	1.15	1.40
France	0.64	0.76	0.42	0.46	0.37	0.36	-0.29**	2.05*	1.92*
Other	0.46	0.38	0.51	0.58	0.49	0.13	0.01	2.48*	1.89*
<i>Allocation Changes</i>									
United States	0.39	0.87	-0.61	-0.35	0.05	-1.11	-0.12**	1.95*	2.27*
Japan	0.98	0.58	0.74	0.69	-0.12	-1.78	-0.25**	4.73**	5.34**
United Kingdom	-0.44	0.02	0.11	0.07	0.98	-0.12	-0.08	-0.43	2.24*
Germany	0.21	-0.24	-0.19	0.17	-0.22	0.30	-0.03	-0.25	0.63
France	0.04	0.62	-0.23	-0.04	-0.18	-0.08	-0.15**	0.35	1.68
Other	0.09	-0.25	0.04	0.53	-0.25	-0.69	-0.01	0.85	0.30

Table 7.

Tests of Alternative Investment Strategies for Different Economic States. Economic states are based on five variables: the Price-Earnings Ratio and Dividend Yield on the MSCI World Index, Industrial Production and Inflation in the G-7 countries and the U.S. Dollar Trade-Weighted Exchange Rate. As before, for each recommendation date (j), we compute two measures. First, the change in equity for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. Second, a Buying Sentiment Measure, $N_{ratio}(i) = \#buys(i)/\#active(i)$, is computed for equity allocations. The number of Houses recommending an increase in equity holdings is given by $\#buys(i)$, while $\#active(i)$ is the number of Houses recommending a change in equity holdings. Then, at each recommendation date (j), the changes in the state variables since the previous recommendation date are ranked. “Low” (“High”) states are defined as those observations where the change in the state variable is below (above) the sample median. In the performance rank cells for the change in equity, we report the grand average across all Houses and all recommendation dates. We report the average $N_{ratio}(i)$ across all recommendation dates in each performance rank cell. At each recommendation date (j), High-Bond (Low-Bond) Houses recommend higher (lower) than the average date (j) bond holdings. We report a t-test and a Wilcoxon Rank-Sum test for the difference in means between the low and high values for the change in equity and $N_{ratio}(i)$. One asterisk (*) or two asterisks (**) denotes significance at the 0.05 or 0.01 level, respectively.

Classification	Value Investing Approach:				Macroeconomic Factor Approach:					
	Change in P/E Ratio		Change in Dividend Yield		G7 Industrial Prod. Growth		G7 Inflation		US \$ Trade Weighted Growth	
	Change		Change		Change		Change		Change	
	in Equity	$N_{ratio}(i)$	in Equity	$N_{ratio}(i)$	in Equity	$N_{ratio}(i)$	in Equity	$N_{ratio}(i)$	in Equity	$N_{ratio}(i)$
<i>Panel A. All Houses.</i>										
Low	-0.60	0.49	0.70	0.59	0.62	0.57	0.54	0.60	-0.41	0.53
High	0.78	0.56	-0.51	0.46	-0.35	0.49	-0.53	0.44	0.49	0.53
t-test	-2.96**	-1.07	2.57**	2.04*	2.07*	1.14	2.17*	2.52*	-1.87	0.01
Rank-Sum Test	-2.10**	-1.07	2.36**	2.13*	2.12*	1.16	2.66**	2.31*	-1.14	0.01
<i>Panel B. High-Bond Houses.</i>										
Low	-1.66	0.46	0.31	0.50	-0.05	0.52	-0.74	0.44	-1.74	0.43
High	0.08	0.45	-1.78	0.42	-1.33	0.40	-0.72	0.48	0.15	0.47
t-test	-2.14*	0.10	2.63**	0.82	1.60	1.28	-0.02	-0.37	-2.34*	-0.41
Rank-Sum Test	-2.12*	0.15	2.31*	0.78	2.19*	1.31	-0.69	-0.35	-1.41	-0.38
<i>Panel C. Low-Bond Houses.</i>										
Low	0.13	0.52	0.99	0.62	1.14	0.57	1.60	0.68	0.71	0.61
High	1.37	0.60	0.46	0.51	0.38	0.55	-0.40	0.41	0.73	0.52
t-test	-2.28*	-1.13	0.96	1.50	1.38	0.34	3.66**	4.11**	-0.04	1.24
Rank-Sum Test	-2.13*	-1.32	1.34	1.69	0.97	0.23	4.25**	3.97**	-0.13	1.17

Table 8.

Tests of Alternative Investment Strategies Using Value Investing Factors. The ranking variables used are either changes in the average Price-Earnings Ratio or changes in the test period average Dividend Yield on the MSCI Country Equity Index between the previous and current recommendation dates. As before, for each recommendation date (j), we compute two measures. First, the change in equity for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. Second, a Buying Sentiment Measure, $N_{ratio}(i) = \#buys(i)/\#active(i)$, is computed for equity allocations. The number of Houses recommending an increase in equity holdings is given by $\#buys(i)$, while $\#active(i)$ is the number of Houses recommending a change in equity holdings. Then, at each recommendation date (j), the changes in the country state variables since the previous recommendation date are ranked from High (1) to Low (5). In the performance rank cells for the change in equity, we report the grand average across all Houses and all recommendation dates. We report the average $N_{ratio}(i)$ across all recommendation dates in each performance rank cell. We report a t-test and a Wilcoxon Rank-Sum test for the difference in means between the High and Low values for the change in equity and $N_{ratio}(i)$. One asterisk (*) or two asterisks (**) denotes significance at the 0.05 or 0.01 level, respectively.

Classification	Change in P/E Ratio		Change in Dividend Yield	
	Change in Equity	$N_{ratio}(i)$	Change in Equity	$N_{ratio}(i)$
<i>Panel A. United States.</i>				
1 (High)	0.01	0.46	-0.77	0.41
2	-0.02	0.49	-0.05	0.48
3	-0.09	0.56	0.30	0.60
4	0.23	0.56	-0.45	0.40
5 (Low)	-0.79	0.34	0.62	0.61
<i>High Versus Low:</i>				
t-test	1.04	1.07	-2.27*	-1.90
Rank-Sum Test	0.16	1.29	-2.10*	-1.71
<i>Panel B. Japan.</i>				
1 (High)	0.43	0.55	-0.59	0.44
2	0.68	0.60	0.19	0.49
3	-0.81	0.36	0.09	0.47
4	-0.68	0.36	0.31	0.64
5 (Low)	-0.59	0.45	0.47	0.54
<i>High Versus Low:</i>				
t-test	2.70**	1.46	-2.29*	-1.02
Rank-Sum Test	2.00*	1.39	-2.12*	-1.13

Table 8, Cont.

Classification	Change in P/E Ratio		Change in Dividend Yield	
	Change in Equity	$N_{ratio}(i)$	Change in Equity	$N_{ratio}(i)$
<i>Panel C. United Kingdom.</i>				
1 (High)	-0.85	0.63	-0.26	0.41
2	-0.12	0.41	0.31	0.56
3	0.35	0.53	0.04	0.47
4	0.09	0.46	0.50	0.54
5 (Low)	-0.02	0.57	-0.32	0.58
<i>High Versus Low:</i>				
t-test	-0.87	0.43	0.14	-1.86
Rank-Sum Test	0.09	0.58	-1.54	-1.74
<i>Panel D. Germany.</i>				
1 (High)	-0.10	0.53	0.15	0.51
2	-0.18	0.43	-0.08	0.52
3	0.11	0.50	-0.20	0.36
4	0.03	0.46	0.24	0.58
5 (Low)	0.23	0.56	0.06	0.57
<i>High Versus Low:</i>				
t-test	-1.61	-0.40	0.40	-0.89
Rank-Sum Test	-0.91	-0.26	-0.44	-0.95
<i>Panel E. France.</i>				
1 (High)	0.16	0.60	0.14	0.48
2	-0.03	0.50	-0.09	0.49
3	0.13	0.52	-0.01	0.50
4	-0.25	0.41	0.10	0.56
5 (Low)	0.20	0.52	0.04	0.54
<i>High Versus Low:</i>				
t-test	-0.33	0.72	0.54	-0.45
Rank-Sum Test	0.41	0.62	-0.83	-0.58

Table 9.

Tests of Alternative Investment Strategies Using Macroeconomic Factors. The ranking variables used are Industrial Production Growth, Inflation, and growth in the \$US Exchange Rate, by country, between the previous and current recommendation dates. As before, for each recommendation date (j), we compute two measures. First, the change in equity for each House is computed as the difference between the current recommended holding and the recommended holding at the previous recommendation date. Second, a Buying Sentiment Measure, $N_{ratio}(i) = \#buys(i)/\#active(i)$, is computed for equity allocations. The number of Houses recommending an increase in equity holdings is given by $\#buys(i)$, while $\#active(i)$ is the number of Houses recommending a change in equity holdings. Then, at each recommendation date (j), the changes in the country state variables since the previous recommendation date are ranked from High (1) to Low (5). In the performance rank cells for the change in equity, we report the grand average across all Houses and all recommendation dates. We report the average $N_{ratio}(i)$ across all recommendation dates in each performance rank cell. We report a t-test and a Wilcoxon Rank-Sum test for the difference in means between the High and Low values for the change in equity and $N_{ratio}(i)$. One asterisk (*) or two asterisks (**) denotes significance at the 0.05 or 0.01 level, respectively.

Classification	Industrial Production Growth		Inflation		US \$ Exchange Rate Growth	
	Change in Equity	$N_{ratio}(i)$	Change in Equity	$N_{ratio}(i)$	Change in Equity	$N_{ratio}(i)$
<i>Panel A. United States.</i>						
1 (High)	0.24	0.51	0.25	0.57	-0.81	0.45
2	-0.03	0.46	-0.38	0.43	0.41	0.48
3	0.05	0.60	-0.34	0.52	-0.61	0.43
4	-0.78	0.42	0.29	0.56	-0.01	0.59
5 (Low)	0.53	0.43	-0.09	0.44	0.12	0.48
<i>High Versus Low:</i>						
t-test	-0.11	0.79	0.53	1.18	-1.04	-0.21
Rank-Sum Test	-0.01	0.47	0.21	0.64	-0.82	-0.05
<i>Panel B. Japan.</i>						
1 (High)	-0.25	0.50	-0.51	0.45	-0.65	0.40
2	0.41	0.46	0.53	0.51	0.68	0.71
3	0.04	0.52	-0.07	0.58	0.00	0.39
4	1.08	0.62	-0.48	0.39	0.45	0.52
5 (Low)	-1.11	0.39	-0.03	0.49	0.10	0.54
<i>High Versus Low:</i>						
t-test	1.74	1.26	-1.01	-0.44	-1.83	-2.18*
Rank-Sum Test	1.16	1.21	-0.75	-0.32	-2.49*	-2.15*

Table 9, Cont.

Classification	Industrial Production Growth		Inflation		US \$ Exchange Rate Growth	
	Change in Equity	$N_{ratio}(i)$	Change in Equity	$N_{ratio}(i)$	Change in Equity	$N_{ratio}(i)$
<i>Panel C. United Kingdom.</i>						
1 (High)	-0.05	0.49	-0.26	0.53	-0.07	0.43
2	-0.11	0.40	0.02	0.47	0.91	0.58
3	-0.03	0.57	0.57	0.52	-0.06	0.43
4	0.17	0.54	-0.16	0.45	0.06	0.58
5 (Low)	0.11	0.51	-0.55	0.48	-0.24	0.56
<i>High Versus Low:</i>						
t-test	-0.83	-0.14	0.54	0.23	0.46	-1.58
Rank-Sum Test	-1.31	-0.37	0.71	0.17	-1.83	-1.48
<i>Panel D. Germany.</i>						
1 (High)	-0.13	0.51	0.32	0.53	0.03	0.58
2	0.19	0.60	-0.50	0.28	0.20	0.53
3	0.29	0.48	0.16	0.52	0.01	0.44
4	-0.05	0.50	-0.05	0.50	-0.11	0.51
5 (Low)	0.01	0.44	0.13	0.57	-0.01	0.51
<i>High Versus Low:</i>						
t-test	-0.75	1.02	0.51	-0.47	0.17	0.97
Rank-Sum Test	0.72	0.99	0.61	-0.41	0.95	0.77
<i>Panel E. France.</i>						
1 (High)	-0.18	0.35	0.01	0.46	0.08	0.44
2	-0.16	0.43	0.16	0.62	0.20	0.57
3	0.14	0.55	-0.17	0.49	-0.10	0.52
4	0.02	0.51	-0.05	0.42	-0.06	0.55
5 (Low)	0.28	0.68	0.27	0.57	-0.09	0.30
<i>High Versus Low:</i>						
t-test	-1.94*	-2.61**	-0.68	-0.61	0.83	0.92
Rank-Sum Test	-2.56**	-2.28*	-1.16	-0.53	0.93	0.88